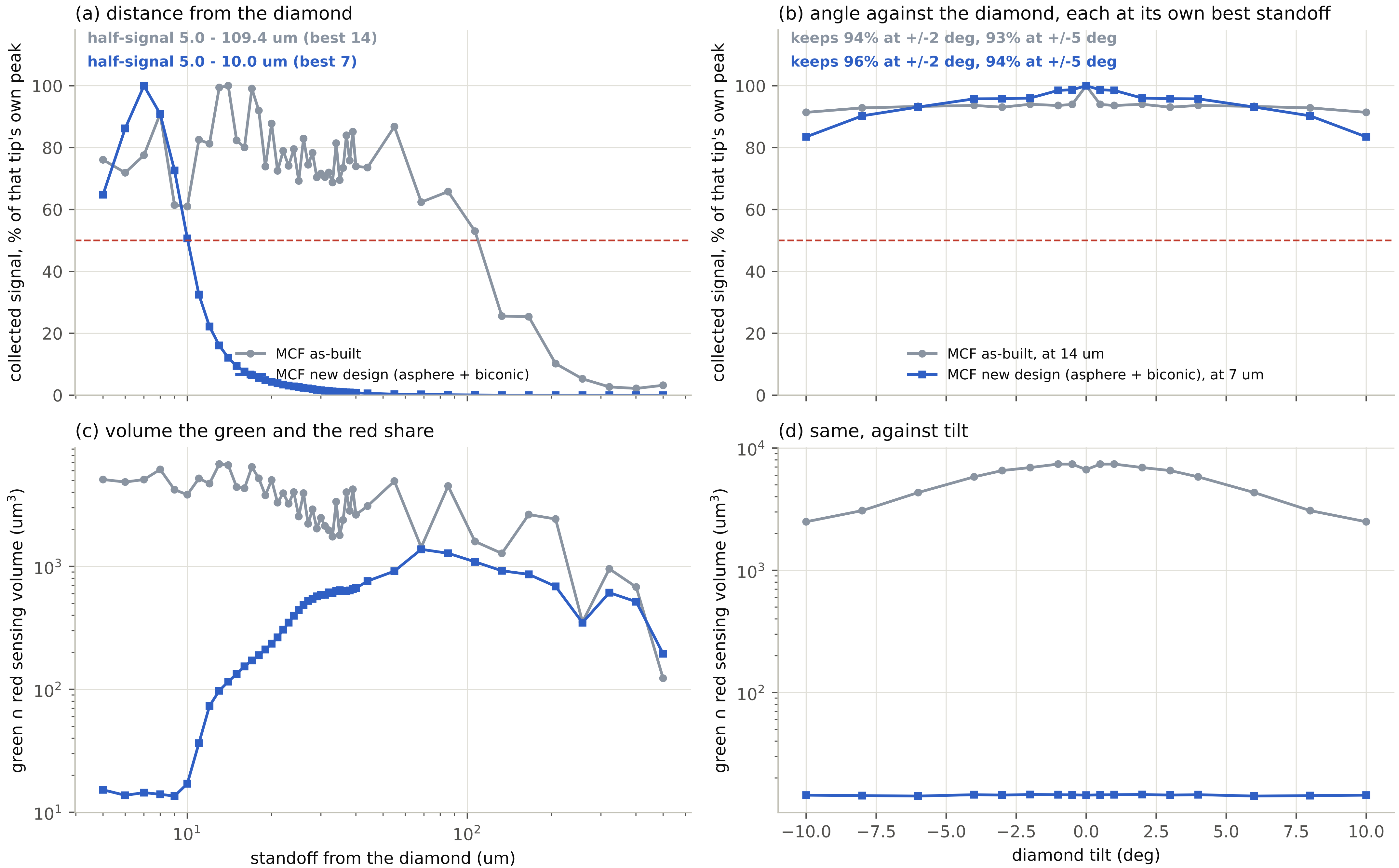

Signal is the product of the traced 532 nm excitation and the six-core collection at every point of the 80-90 um layer, so an NV counts only where the green reaches it and a core can see it; both are re-traced at every standoff and every tilt. (a,b) are normalised to each tip's own peak and compare how fast a tip loses light, not how much it collects -- at its own optimum the new design collects about 426x the as-built tip. Dashed line is half signal. (c,d) give the participation-ratio volume of that product, threshold-free so it stays smooth off-optimum; growing means a blurrier probe, not a better one. Tilt rotates the diamond plane about the fiber y-axis, with all six side cores traced separately.
The few-percent ripple on the as-built curve reproduces under grid refinement, so it is the model's ray optics -- hard apertures and no wave averaging -- rather than numerical noise.